Week 10: Payment Rails and Dollar Dominance

FINA3020 at CUHK Business School

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Assume a Hong Kong firm must pay USD 1 million to a foreign supplier. We will study

- How a payment moves money between bank ledgers
- Why cross-border payments are slow and costly, and why most payments use US dollars
- New rails: linked fast payment systems, CIPS, mBridge, and stablecoins
- Decisions for corporate treasurers and investors (game)
- Dollar dominance: invoicing, funding, safe assets, and the US external balance sheet

A Payment Moves a Claim between Ledgers [Core Concept]

A **ledger** records account balances and transactions. A bank deposit is a claim on (liability of) a bank. A payment changes who holds the claim.

- **Same bank**: the bank decreases the payer's balance and increases the payee's balance.
- **Different banks**: the payer's bank must pay the payee's bank. The two banks settle with a transfer of **reserves** on the central bank's ledger.

Example: a payer at Bank A pays HKD 100 to a payee at Bank B.

Ledger	Liabilities	Assets
Bank A	Payer's deposit -100	Reserves at the central bank -100
Central bank	Bank A's reserve account -100	
	Bank B's reserve account $+100$	
Bank B	Payee's deposit $+100$	Reserves at the central bank $+100$

Reserves are the **settlement asset**. They are a claim on the central bank, with no credit risk.

Payment Rails: Authorization, Clearing, and Settlement [Core Concept]

A **payment rail** is the set of rules, messages, and ledgers that moves money from a payer to a payee. Examples: bank transfers and wires, card networks such as Visa and Mastercard, instant payment systems, and mobile wallets.

Each non-cash rail has three stages:

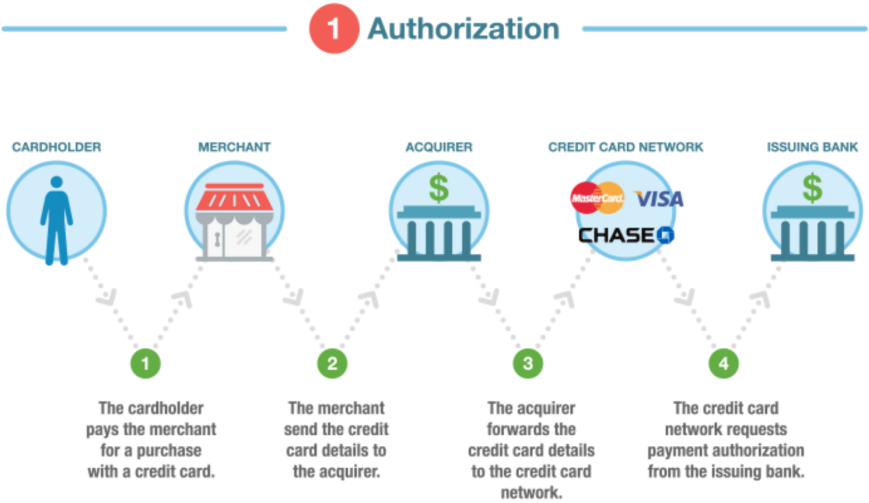
- **Authorization**: the payer's bank checks the payer and the available funds, then approves the payment.
- **Clearing**: the banks exchange and match payment messages, then calculate the amount each bank owes. **Netting** offsets amounts owed in both directions.
- **Settlement**: the banks transfer the settlement asset. **Final** settlement is irrevocable.

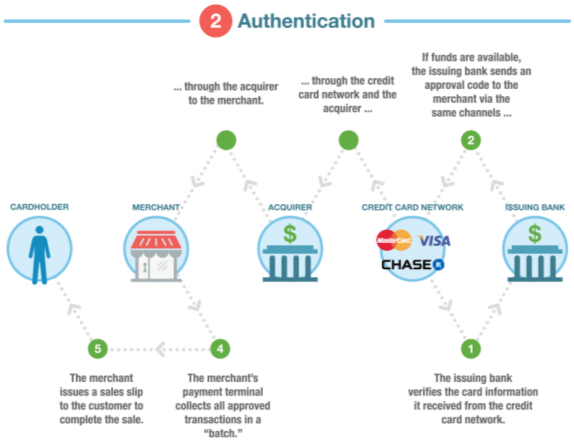
A wire clears and settles each payment at once. A card payment gets authorization in seconds, then banks settle one or two business days later.

Card Payments: Authorization through Settlement

- **Authorization**: the merchant's bank (the **acquirer**) sends the request through the card network to the cardholder's bank (the **issuer**). The issuer approves and holds the amount on the card.
- **Authentication** checks the identity of the cardholder, for example with a PIN or a one-time code.
- **Clearing**: the network matches the day's transactions and calculates the net amount for each bank.
- **Settlement**: the issuer pays the network's settlement bank, which pays the acquirer. The acquirer credits the merchant, less the merchant fee.
- For a foreign card, the network or the issuer converts the currency. It adds a spread and a cross-border fee.

Card Networks Illustration: Authorization





The figure labels checks and approval "authentication. Approval is extbfauthorization; extbfcapture records approved payments for clearing.


3 Clearing & Settlement
The issuing bank posts the transaction to the cardholder's account, and the cardholder pays the bill.
7
The acquirer credits the merchant less a "merchant discount rate."
6
The credit card network pays the acquirer its percentage of the remaining funds.
5
The issuing bank transfers the funds to the acquirer less an "interchange fee."
4
CARDHOLDER
MERCHANT
ACQUIRER
CREDIT CARD NETWORK
ISSUING BANK
MasterCard
VISA
CHASE
1
The merchant sends the batch of approved transactions to the acquirer.
2
The acquirer forwards the batch to the credit card network for settlement.
3
The credit card network sends each approved transaction to the appropriate issuing bank.

Domestic Rails: Wires, ACH, and Instant Payments

[Background Context]

- **Wires (real-time gross settlement)**: each payment settles separately and immediately, in central-bank money. Examples: Fedwire in the US and Hong Kong's HKD Clearing House Automated Transfer System (CHATS). Firms use wires for large, urgent payments.
- **ACH (automated clearing house)**: the system collects many small payments, nets them, and settles the net amounts in batches. Firms use ACH for payroll and bills.
- **Instant payment systems**: the payee receives usable funds in seconds, 24 hours a day. Examples: FedNow and Real-Time Payments (RTP) in the US; the Faster Payment System (FPS) in Hong Kong.
- Distinguish the interface (the banking app) from the rail. The rail sets the speed, the cost, and the time of finality.
- An instant payment is final at once. A fraudulent or incorrect payment is difficult to recover.

US payment-system institutions →

Domestic Fast Payment Systems

Many economies run instant payment systems: Brazil Pix, Hong Kong FPS, India's Unified Payments Interface (UPI), and US FedNow. A central bank, a public utility, or a bank consortium operates each system.

Common design features:

- *Always on*: customers can pay 24 hours a day, 7 days a week.
- *Fast*: the payee receives usable funds in seconds.
- *Broad reach*: all participating banks and wallets use one scheme, so any user can pay any other user.
- *Simple addressing*: users pay to a phone number, an alias, or a QR code.
- *Low cost*: transfers between people are often free. Merchant fees are low compared with card fees.

These features apply to payments within one currency area. Cross-border payments are slower and more expensive.

Country systems in detail → Pix, FPS, and UPI designs →

Why Cross-Border Payments Are Slow and Costly

[Core Concept]

No central bank keeps accounts for all the world's banks. A Hong Kong bank without a US branch has no account at the Federal Reserve. To pay in USD, it holds a USD deposit at a US bank.

- **More banks**: a payment often moves through two to four banks. Each bank checks the payment, charges a fee, and can hold the payment.
- **Currency conversion**: a bank or dealer converts at a rate that includes a spread.
- **Time zones and cut-off times**: the settlement systems of the two currencies open at different hours.
- **Compliance checks**: each bank screens the payment for sanctions and money laundering. A query can stop a payment for days.

G20 retail-payment targets for end-2027: 75% available within one hour, and a global average cost of at most 1% (remittances have a separate cost target).

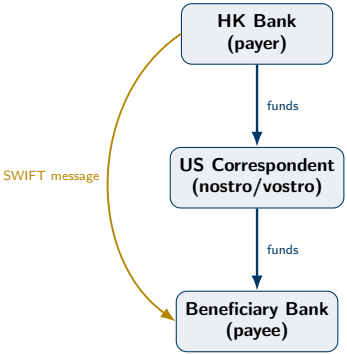
Source: FSB, Targets for Addressing the Four Challenges of Cross-Border Payments (2021).

Correspondent Banking: Nostro and Vostro Accounts

[Core Concept]

- A **correspondent bank** holds accounts for other banks in its own currency.
- **Nostro** ("ours"): our account at your bank. **Vostro** ("yours"): your account at our bank. One account is a nostro on one bank's books and a vostro on the other bank's books.
- **SWIFT** sends the payment instruction between banks. The correspondent debits one account and credits another.
- The paying bank keeps a **prefunded** balance (cash placed in advance) in its nostro or uses a credit line from the correspondent.
- Each bank in the chain charges a fee and can add delay.

Nostro and vostro T-accounts →



Example: A Hong Kong Importer Pays a Thai Supplier in USD [Core Concept]

A Hong Kong importer pays USD 100,000 to a supplier in Bangkok at spot rate 7.80 HKD per USD. HK Bank and Bangkok Bank each hold a USD account at a different US bank.

Step	Ledger	Entry
1	HK Bank, Hong Kong	Importer's HKD deposit −780,000. HK Bank sends a SWIFT instruction.
2	US Bank 1, US	HK Bank's USD account (HK Bank's nostro) −100,000
3	Fedwire or CHIPS	US Bank 1 pays US Bank 2 USD 100,000
4	US Bank 2, US	Bangkok Bank's USD account +100,000
5	Bangkok Bank, Thailand	Supplier's USD account +100,000, or THB at Bangkok Bank's rate

The USD funds stay in US banks. Steps 2 to 4 take place on US bank ledgers during US business hours. Most USD payments between non-US banks settle in this way.

The Full Cost of a Cross-Border Payment

[Core Concept]

$$\text{Total cost} = \text{fees} + \text{FX conversion cost} + \text{committed-cash cost} + \text{risk charge}$$

- **Fees**: each bank in the chain can charge. Some banks deduct their fee from the payment, so the payee receives less than the invoice amount.
- **FX conversion cost**: the amount times the spread between the quoted rate and the mid-market rate.
- **Committed-cash cost**: prefunded cash has an opportunity cost; credit uses funding capacity. For cash held, cost = balance $\times$ interest rate $\times$ days/360.
- **Risk charge**: the price of the risk that a counterparty fails after the firm or its bank has paid.

When the payee has a deadline, add the expected cost of delay. Compare routes for the same amount received and the same delivery date.

Separate three-route illustration → Five-day cost scenario →

Settlement Risk and Payment versus Payment [Core Concept]

An FX trade has two payments, one in each currency. If the two payments settle at different times, one bank pays first.

- **Herstatt, 1974**: counterparties paid Deutsche marks to Bankhaus Herstatt. They expected US dollars later that day in New York. German regulators closed the bank first, and its New York correspondent stopped the dollar payments.
- The full principal was exposed when the dollar payments failed. This risk is **settlement risk**, also called **Herstatt risk**.
- **Payment versus payment (PvP)**: each currency leg settles only if the other leg settles. **CLS** (Continuous Linked Settlement) settles FX trades in 18 currencies by PvP.
- PvP removes principal risk. Each bank must still deliver its currency on time.

Comparison: Delivery versus payment (DvP) links securities delivery to the matching cash payment. Payment versus payment (PvP) links the two currency legs of an FX trade.

Source: BIS, Settlement Risk in Foreign Exchange Transactions (1996); CLS, CLSSettlement.

SWIFT Messages and Dollar Settlement Systems

[Background Context]

- **SWIFT** is a messaging network owned by its member banks. It carries standard payment instructions. It does not hold accounts or move money.
- **SWIFT gpi** tracks each payment from bank to bank. The payer can see which bank holds the payment and which fees each bank deducted.
- **Fedwire** (Federal Reserve) settles each USD payment at once, in reserves. **CHIPS** (Clearing House Interbank Payments System, owned by large banks) nets payments to save liquidity. These are the main US systems for large-value interbank USD payments.
- **USD CHATS** in Hong Kong settles USD payments during Asian business hours, with HSBC as settlement institution. It links to the HKD, EUR, and CNH systems in Hong Kong, so FX trades in those pairs settle PvP.

Source: HKMA, Payment Systems.

Why Most Cross-Border Payments Use the Dollar

[Core Concept]

A Thai bank that pays in KRW often converts THB to USD, then USD to KRW. The USD is a **vehicle currency**.

- **Fewer markets**: with N currencies, direct trading needs $N(N-1)/2$ currency pairs. A vehicle currency needs $N-1$. For 20 currencies: 190 pairs versus 19.
- **Narrower spreads**: trading concentrates in USD pairs, so their spreads are the narrowest. Two narrow spreads can cost less than one wide spread.
- **Fewer accounts**: one USD nostro lets a bank pay almost any counterparty. Each nostro is a prefunded balance.
- **Network effect**: when more banks and firms use the dollar, its cost decreases. The lower cost attracts more users.

In April 2025, the USD was on one side of 89% of all FX trades. The euro was on one side of 29%.

Source: BIS Triennial Central Bank Survey, OTC Foreign Exchange Turnover in April 2025.

Example: A Thai Importer Pays a Korean Supplier [Core Concept]

A Thai importer pays a Korean supplier the KRW equivalent of USD 1,000,000. Hypothetical dealer spreads are measured from the mid-market rate. Cost = amount × total spread.

Route	Spread	Cost (USD)
Direct: THB → KRW	0.50%	5,000
Vehicle: THB → USD → KRW	0.10% + 0.10%	2,000

- The route through the dollar uses two conversions and saves USD 3,000.
- The Thai bank uses its USD nostro in New York. A direct route needs a prefunded KRW nostro in Seoul.
- On both routes, the importer pays THB and the supplier receives KRW. Only the banks use USD.

Think: What must change for the direct THB → KRW route to become cheaper?

New Payment Rails: Ledger and Currency [Core Concept]

For each new rail, identify the settlement ledger and the currency.

Approach	Example	Ledger that settles	Currency
Link domestic fast payment systems	Payment Connect, Nexus	Each country's own system	Local currencies
Clearing system for another currency	CIPS (Cross-Border Interbank Payment System)	CIPS accounts of participant banks	CNY
Shared central-bank platform	mBridge	One ledger for several central banks	Central-bank digital currencies (CBDCs)
Tokens on a blockchain	USDT, USDC (USD tokens)	Blockchain: a shared transaction ledger; issuer holds reserves	Over 99% USD by value

New rails can shorten correspondent chains while retaining the currency. Stablecoins often move USD on a new ledger.

Source: Stablecoin share: BIS Annual Economic Report 2025, Chapter III.

Mainland & HK Payment Connect

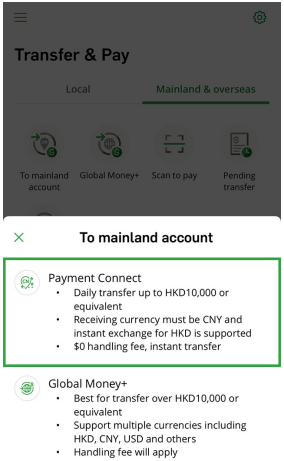
[Background Context]

Payment Connect launched on 22 June 2025. It links mainland China's Internet Banking Payment System (IBPS) with Hong Kong's FPS.

Residents of both places can send real-time cross-boundary remittances through participating institutions, within set rules and limits.

Merchant QR payments across the boundary use separate schemes.

Source: HKMA and PBoC, launch announcement, 20 June 2025.



Project Nexus: One Hub for Fast Payment Systems

[Background Context]

- A bilateral link such as Payment Connect joins two systems. With N systems, bilateral links need $N(N-1)/2$ connections, the same number as direct currency pairs.
- **Project Nexus** designs one common platform. Each domestic fast payment system makes one connection and reaches all other members.
- In 2025, the central banks of India, Indonesia, Malaysia, the Philippines, Singapore, and Thailand formed Nexus Global Payments to operate it. The BIS transferred the project to this organization.
- **Project Aperta** uses the same hub design for open finance. It shares permitted account data and does not settle payments.

Source: BIS, Project Nexus; BIS, Project Aperta.

Who Keeps the Ledger?

[Core Concept]

Every payment system keeps a **ledger**: a record of balances and transactions. The ledger must stop payments without funds and stop the same funds from being spent twice.

Centralized ledger

- One operator keeps the record: a commercial bank, a central bank, or a card network.
- Fast and cheap at large scale.
- Users trust the operator, its controls, and its supervisor.

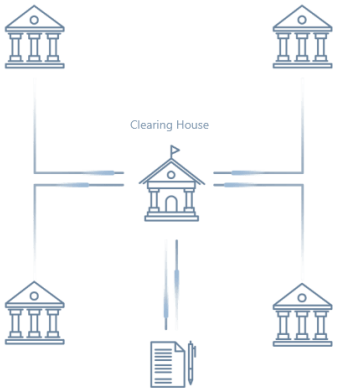
Distributed ledger

- Many validators keep copies and agree on each update through a protocol, with **proof of work** (computing effort) or **proof of stake** (assets pledged by validators).
- **Validators** check and approve transactions. A permissioned ledger restricts who can validate; a public, permissionless blockchain allows open participation.
- Users trust the protocol, the validators, and the code.

Ledger economics trilemma →

Centralized and Distributed Ledger Technology

[Background Context]



DLT means distributed ledger technology. Its governance can still be centralized, with approved validators only.

Centralized vs Distributed Ledgers, Continued



CBDCs, Tokenized Deposits, and Stablecoins

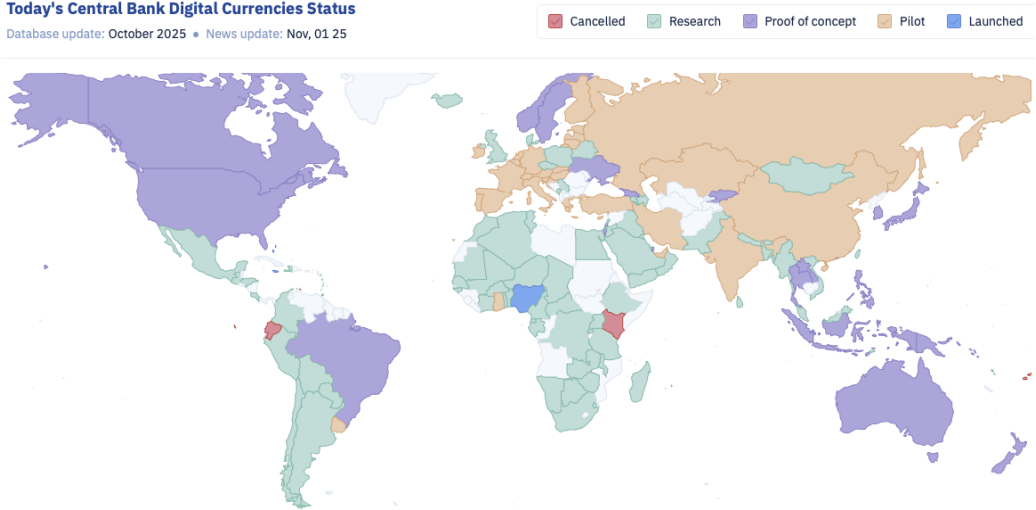
[Core Concept]

- A **central bank digital currency (CBDC)** is a digital liability of a central bank. A **wholesale** CBDC settles payments between banks. Households and firms hold a **retail** CBDC.
- A **token** is a digital record of value or a claim. A **tokenized deposit** records a bank deposit on such a ledger. It is still a liability of the commercial bank.
- A **reserve-backed stablecoin** is a token issued by a private entity against reserve assets. Redemption rights depend on the issuer's terms.
- A **smart contract** is code on the ledger that moves funds when set conditions hold. Example: payment on receipt of a verified shipping document.

For each token, ask: who owes the holder, what backs the claim, and where does final settlement take place?

Shipment escrow example →

CBDC Status around the World (October 2025)



Source: CBDC Tracker.

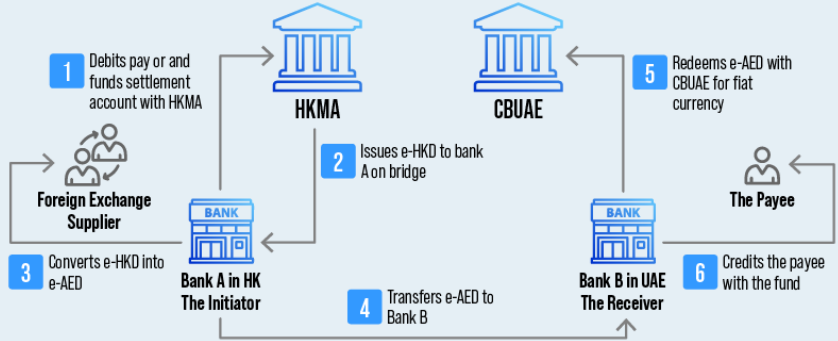
mBridge: One Ledger for Several Central Banks

- Each member central bank issues a wholesale CBDC on one shared ledger. Members: the People's Bank of China, the HKMA, the Bank of Thailand, the Central Bank of the UAE, and the Saudi Central Bank (from 2024).
- Commercial banks on the platform pay each other directly across borders. FX trades settle PvP on the platform, without a correspondent chain.
- A **liquidity-saving mechanism** queues payments in both directions and settles the net amounts. Banks keep smaller prefunded balances.
- mBridge reached the minimum viable product stage in 2024. The BIS transferred the project to the member central banks in October 2024.

Dunbar, Icebreaker, and mBridge →

Wholesale CBDC cross-border payment of a transaction from Hong Kong to UAE

Figure 1. The procedure of cross-border payments on bridge



Source: TABInsights

mBridge lowers funding costs and alleviates liquidity constraints on banks through liquidity saving mechanism (LSM)

Figure 2. Traditional Settlements versus LSM-enabled Settlements

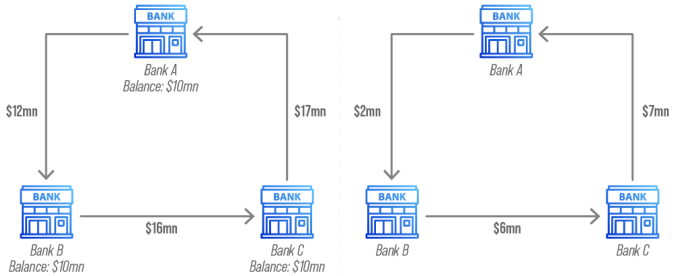


Figure 2a. Traditional Settlements

Figure 2b. LSM-enabled Settlements

Source: TABInsights

The mechanism offsets queued payments in both directions. Banks fund only the net amounts.

Stablecoin "Sandwich" as a Cross-Border Payment Rail [Core Concept]

- **Three legs**: the sender buys USD tokens with local money (on-ramp). The tokens move on a blockchain. The recipient sells the tokens for local money (off-ramp).
- The token leg settles in minutes, 24 hours a day. The two conversion legs use banks and bank hours.
- **Total cost**: two conversion spreads, the blockchain fee, and the exchange or redemption fee.
- Large issuers hold most reserves in short-term US Treasury bills, repos, and bank deposits. Only verified customers can redeem directly with the issuer. Other holders sell on an exchange.
- Over 99% of stablecoin value is in USD tokens (BIS, 2025), so this route increases the use of the dollar.

Source: BIS Annual Economic Report 2025, Chapter III.

Wallets and stablecoins: legal claims →

Smart Contracts and Atomic FX Settlement

[Core Concept]

Duffie and Wang (2026) model a payment that uses several dealers in sequence, for example THB → USD stablecoin → KRW.

- Each dealer trades one leg. When the dealers agree the legs one at a time, a dealer that has completed leg 1 is exposed on leg 2.
- The next dealer can then demand better terms on leg 2. This is **holdup**.
- A smart contract can execute all legs together or none of them. This is **atomic settlement**.
- Atomic settlement removes principal risk and holdup for the legs that the contract controls. Liquidity, market, and legal risks remain.

Atomic FX bargaining detail →

Example: The Multinational Treasurer's Payment Calendar

A Hong Kong parent sells in Europe, buys components in the US, and pays staff in Hong Kong. Treasury must have funds in the right account when each payment is due.

Date	Legal entity	Cash flow	Amount
Monday	European subsidiary	Customer receipts	EUR 500,000
Wednesday	Hong Kong parent	Supplier payment	USD 300,000
Friday	Hong Kong parent	Payroll	HKD 1,200,000

- The group may have surplus cash while one subsidiary lacks the currency it needs.
- Cash becomes available at different times as customers pay and transfers settle.

Netting Example

[Core Concept]

Two subsidiaries have reciprocal invoices due on the same date: the European entity owes the Hong Kong entity EUR 500,000, and Hong Kong owes Europe EUR 320,000.

$$\text{Net payment from Europe to Hong Kong} = 500{,}000 - 320{,}000 = \text{EUR } 180{,}000.$$

- If their agreements and local rules permit netting, treasury pays one net amount and records both invoices as paid.
- At an assumed 0.4% transfer cost, two gross transfers cost EUR 3,280. One net transfer costs EUR 720, saving EUR 2,560.
- If Hong Kong keeps the EUR 180,000 to pay a later euro supplier bill, it avoids two FX conversions.

What a Payment Provider Does

[Background Context]

- A provider collects the payer's money, converts it if necessary, and delivers it to the beneficiary. For a local payout, the provider uses a partner bank or payment system in the destination country.
- **Wise Business**: a firm can pay many recipients in one batch, funded from its Wise balance or by bank transfer.
- **Airwallex**: a firm can pay out through local rails or SWIFT. The required details, currencies, and fees depend on the route.
- Treasury compares the total cost, the amount the recipient receives, and the arrival date.

Source: Wise, funding batch payments.

Source: Airwallex, payout network.

Using a Provider for a Supplier Payment

[Background Context]

The supplier must receive exactly USD 100,000 by Wednesday. Treasury uses HKD cash.

1. Confirm the invoice, beneficiary name, account currency, bank details, and payment reference. Verify changes to the supplier's bank details through a known contact.
2. Request a quote with the **recipient amount fixed**. Record HKD debited, all fees, quote expiry, and the stated delivery window.
3. Approve the transfer and send funds to the provider before the cut-off.
4. Track completion and reconcile the HKD debit, FX rate, fees, and supplier receipt to the invoice.

Provider Comparison: A Fixed Recipient Amount

[Background Context]

Hypothetical quotes for delivering USD 100,000. The mid-market rate is 7.80 HKD per USD. Each FX markup is a percentage increase in that rate. The sender pays all fees separately.

	Bank A	Provider B	Provider C
FX markup	0.60%	0.25%	0.10%
Sender fee (HKD)	250	80	50
Additional delivery fee (HKD)	150	0	0
Stated arrival	1 business day	1 business day	3 business days

1. Calculate the HKD debit for each route.
2. The supplier's deadline is two business days away. Select the cheapest route that meets the deadline.

Provider Comparison: Cost and the Deadline

$$\text{HKD debit} = 100{,}000(7.80)(1 + \text{markup}) + \text{fees}.$$

Route	Conversion amount	Total HKD debit	Meets deadline?
Bank A	784,680	785,080	Yes
Provider B	781,950	782,030	Yes
Provider C	780,780	780,830	No

- B costs HKD 3,050 less than A and meets the deadline. C arrives too late.
- Sometimes it is unclear to the payer how much the intermediary charges in fees, so the recipient does not receive the full payment.

Delay Risk Can Change the Route Choice

[Core Concept]

Now assume A is late with probability 0.5% and B with probability 5%. A late delivery causes a known HKD 100,000 production loss. Otherwise the production loss is zero.

	Bank A	Provider B
Certain payment cost	785,080	782,030
Expected delay loss	$0.005(100{,}000) = 500$	$0.05(100{,}000) = 5{,}000$
Expected total cost	785,580	787,030

A now has the lower expected cost. B is cheaper when the loss per late payment is below

$$L^* = \frac{785{,}080 - 782{,}030}{0.05 - 0.005} = \text{HKD } 67{,}777.78.$$

A Cash Calendar for an International Fund

Hypothetical USD cash flows, in millions. Initial available cash is 1.0. Assume no overdraft or intraday credit.

Date	Transaction	Cash flow	Running cash
Today	FX hedge collateral	−0.8	0.2
Tomorrow	New bond purchase settles	−2.0	−1.8
Day 2	Previous bond sale settles	+3.0	1.2

1. How much additional cash must arrive before tomorrow's purchase?
2. Compare an earlier FX conversion, committed credit, and postponing the purchase.

Payment Route Exercise

[Core Concept]

Pay a supplier **USD 1,000,000 by Wednesday**, before the customer pays on Thursday. Start with USD 400,000; payroll of USD 300,000 is due Friday.

1. Understand the cash ledger, read four examples, and practise five calculations.
2. For each market, select **correspondent banks or a stablecoin bridge**, and a **USD 600,000 or USD 1,000,000 credit line**.
3. Play the emerging-market week and respond when the customer pays late.
4. Compare expected costs across possible weeks. Reflect and submit; wait for a receipt.

25–30 minutes. Any plan can earn participation credit. Hints and solutions are free.

Examples | Treasury workshop

Open exercise

Student explanation

Expected weekly cost (USD). Hypothetical model; small-line plans buy the urgent increase when customer funds are held. Fees and interest are tallied separately from the cash ledger.

Monday plan	Developed-market partners	Emerging-market partners
Correspondent banks + small line	**603**	1,475
Correspondent banks + large line	753	1,425
Stablecoin bridge + small line	853	1,235
Stablecoin bridge + large line	1,003	**1,185**

- Developed markets: both routes pay on Wednesday. The bank route has the lower fee.
- Emerging markets: the stablecoin route saves two days of credit. The large line saves an expected \$50: $0.25(1{,}000) - 200$.
- Neither tool insures nonpayment. Separate expected invoice losses are \$600 and \$6,000; the table excludes them.

Cost tables and ledger explorer

Dollar Dominance Has Several Dimensions [Core Concept]

The first half of the lecture showed the role of the dollar in cross-border payments. The dollar also has other roles. A **safe asset** is expected to preserve value and remain liquid in stress.

Function	Why it matters to a firm or investor
Trade invoicing	Sets the currency of prices and receivables.
Payments and FX trading	USD pairs are the deepest FX markets, so banks often use USD to pay between two other currencies.
International borrowing	Borrowers need the currency to service debt and refinance.
Reserves and collateral	Investors value assets they can sell or pledge during stress.

These roles support each other. A firm that invoices in USD also prefers to borrow and pay in USD. A new payment rail changes how money moves. The contract still sets the currency that the buyer owes.

Source: Federal Reserve, international role of the US dollar (2025).

Five Stages of an International Transaction

Liu divides an international transaction into five stages. Each stage can use a different currency.

Stage	Firm's decision
Pricing	Which currency fixes the invoice and the amount owed?
Financing	Which currency does the buyer borrow before paying?
Risk transfer	Which currency exposures are hedged or insured?
Payment	Which asset and which rail settle the payment?
Reinvestment	Does the recipient hold, invest or convert the proceeds?

Example: a USD stablecoin route changes only the payment stage. The invoice stays in USD. Use of a currency at one stage increases demand for it at the other stages.

Source: Zongyuan Zoe Liu, Foreign Affairs article supplied for Week 10, printed p. 7.

Invoicing Currency and Settlement Currency

[Background Context]

- A contract can set the price at **USD 100,000**. A bank can convert the payment at settlement, so the Chinese exporter receives CNY.
- The buyer still pays USD 100,000. The exchange rate on the conversion date sets the exporter's CNY receipt, so the exporter bears the FX risk.
- 6.5% of China's exports were invoiced in CNY in 2023, and **nearly 30% of China's goods trade was settled in CNY**.
- The two numbers measure different things: exports versus all goods trade, and invoicing versus settlement. Compare each number only with data that use the same measure.

Source: Liu, supplied article, printed p. 8; citing IMF invoicing estimates and Chinese settlement statistics.

Payment Infrastructure and Trade Financing

[Background Context]

- **Payment reach:** 1,829 CIPS participants in June 2026. Through them, CIPS can send payments to more than 5,200 banks in 191 countries.
- **Payment value:** in 2025, CHIPS processed about 19 times the value of CIPS payments. T2, the euro system, processed about 14 times. These ratios compare systems that cover different transactions.
- **Trade finance:** in May 2026, CNY had a 7% share of the value of trade-finance messages on SWIFT. USD had 82.5%. This measure covers trade finance on SWIFT only.

Source: Liu, supplied article, printed pp. 7 and 10–11. All figures are as reported in the article.

Reinvesting Payment Proceeds

[Background Context]

- A firm that receives CNY can keep a deposit, buy an asset, pay a supplier, or convert the CNY. International use of CNY increases when firms keep and invest it.
- Stock Connect and Bond Connect let foreign investors buy mainland stocks and bonds. Swap Connect lets them hedge CNY interest-rate risk. Demand depends on access, liquidity, and the freedom to move funds. CNY green bonds are another asset for these investors. Link to Week 12: for a green bond, examine the issuer's cash flows, who bears losses, liquidity, and currency risk, in addition to the use of proceeds.

Source: Liu, supplied article, printed pp. 11–15.

Dollar Invoicing and Trade-Price Pass-Through

- Under **dominant-currency pricing**, firms in two non-US countries often set sticky prices in USD. **Pass-through** measures how much an exchange-rate change alters an import price.
- If the importer's currency weakens against USD and the dollar invoice stays fixed, the importer's local-currency cost increases. Sticky dollar prices make this pass-through larger.
- For this reason, the exchange rate between the two trading countries explains trade prices less well than the dollar exchange rate. Firms that borrow USD for working capital also have a reason to invoice in USD.
- Currency shares of FX turnover add up to 200%, because each FX trade has two currencies.

Dollar invoice example →

Dollar Pricing and Expenditure Switching

[Core Concept]

Textbook view (producer-currency pricing): each exporter sets a sticky price in its own currency, so the **terms of trade** (export prices relative to import prices) move one-for-one with the exchange rate.

Dominant-currency pricing: most exporters set sticky prices in dollars. Gita Gopinath estimates that close to 90% of China's exports to the US are priced in dollars, so the terms of trade move little.

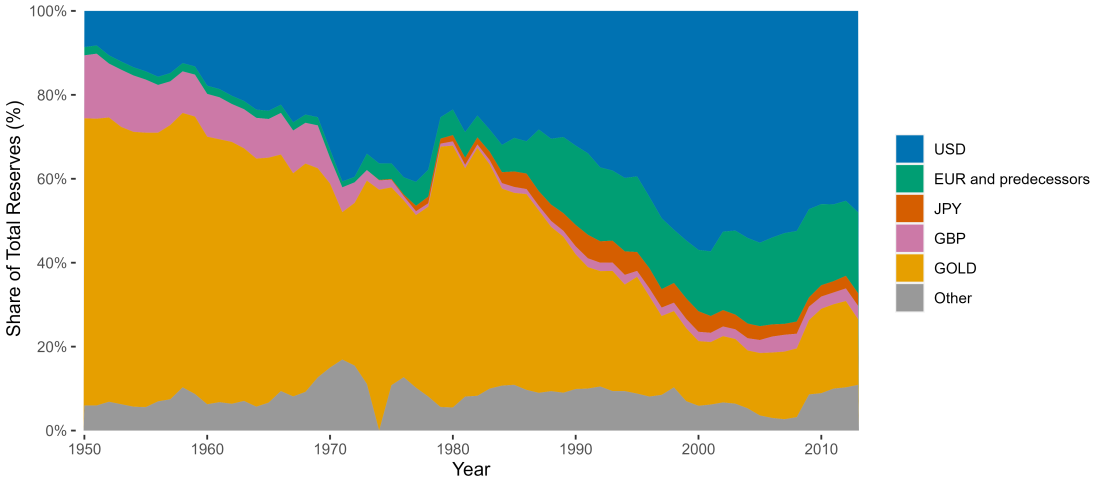
Shock	That country's exports	That country's imports
Dollar depreciates (US)	Rise: cheaper in foreign currency	Little change: dollar price fixed
CNY depreciates (China)	Little change: dollar price fixed	Fall: cost more in CNY

Expenditure switching: buyers shift spending toward goods that became relatively cheaper. Under dollar pricing it works through US exports but not US imports, and through other countries' imports but not their exports.

Source: Gita Gopinath, Conversations with Tyler, Ep. 291, recorded 27 August 2026.

Why prices stay in dollars → Depreciation vs. Tariff →

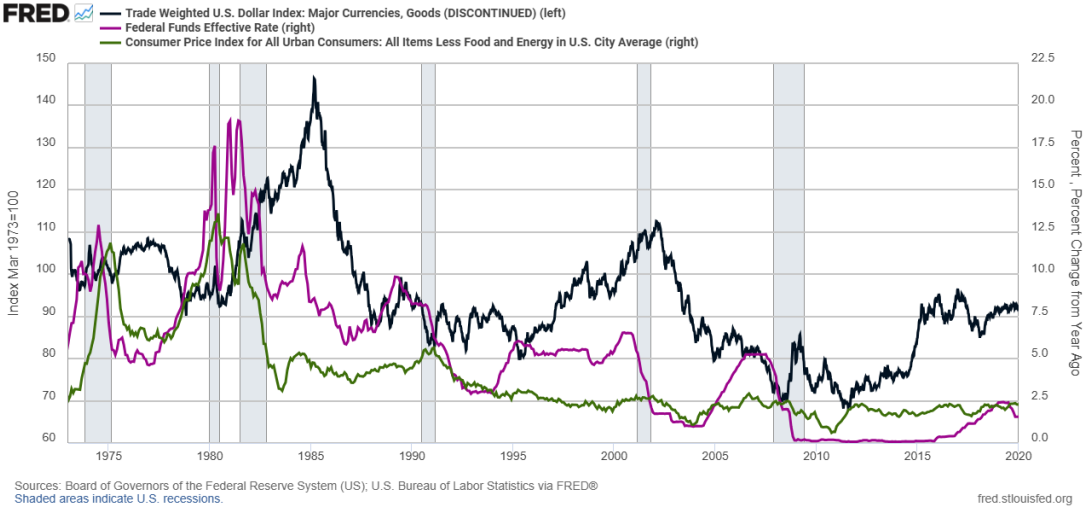
Time Series of the Currency Composition of Reserves



Source: My figure, data from Eichengreen, Mehl, Chitu (2017).

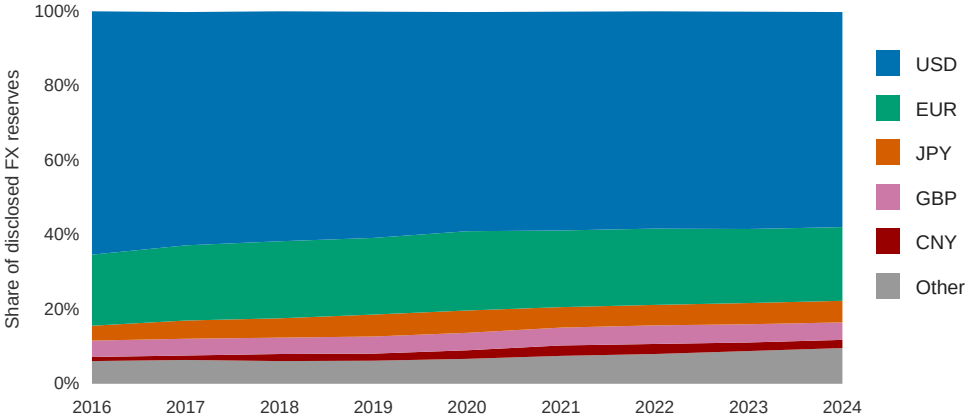
Trade-Weighted Strength of USD and US Monetary Policy

[Background Context]



Official Reserve Currency Shares, 2016–2024

[Background Context]



Published annual data, rounded to one decimal; totals can differ slightly from 100%.

Source: IMF COFER, annual shares of globally disclosed FX reserves, as published in Federal Reserve (2025), Figure 2. Published values are rounded.

The US External Balance Sheet

[Core Concept]

The US international investment position records residents' claims on the rest of the world and foreign residents' claims on the US.

- External assets include foreign direct investment, portfolio securities, loans, and deposits. Many assets have foreign-currency exposure.
- External liabilities include US equity and direct investment as well as Treasury securities, deposits, and other debt. Much US external debt is denominated in USD.
- Changes in asset prices and exchange rates alter both sides. A dollar decline raises the USD value of a fixed foreign-currency claim.
- The US external balance sheet combines the foreign assets and liabilities of households, firms, banks, and government.

Source: BEA, International Economic Accounts: Concepts and Methods.

Example: A Weaker Dollar and US Net Foreign Assets

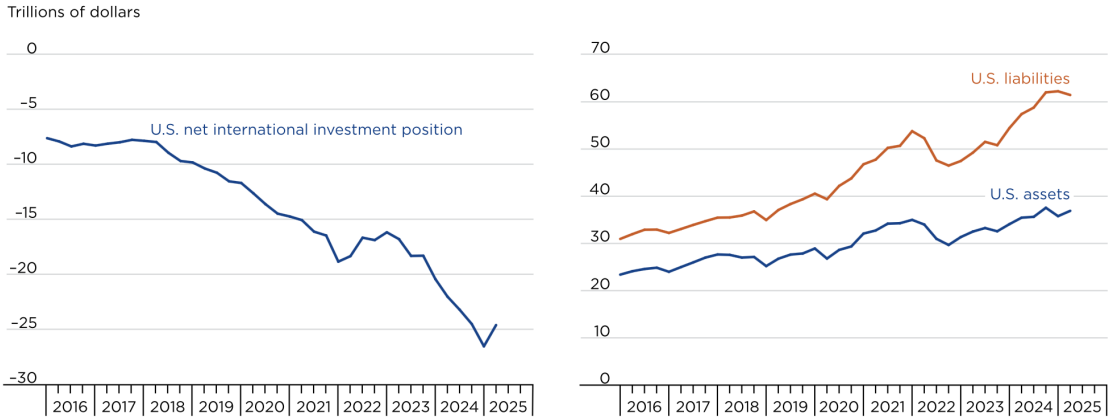
Consider US residents who own EUR 100B of foreign assets and owe USD 100B. Hold the euro asset price and dollar debt fixed.

USD billions	Initially	After the dollar weakens
Exchange rate (USD/EUR)	1.00	1.20
Foreign assets in USD	100	120
Dollar liabilities	100	100
Net foreign assets	0	20

Discuss: Who gains from the exchange-rate change? What happens to a euro investor who holds the USD 100B claim?

US External Position: Historical Evidence

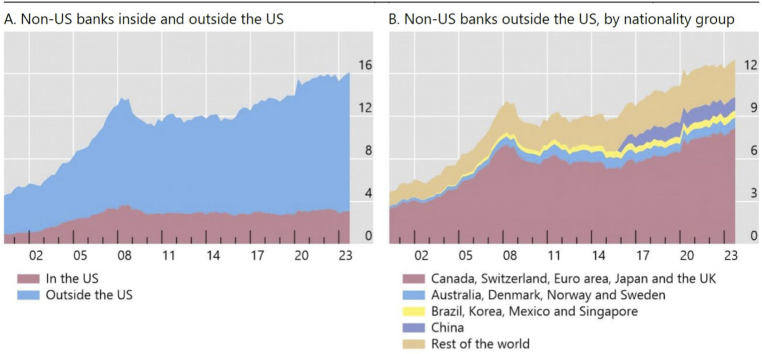
[Background Context]



U.S. Bureau of Economic Analysis

Source: BEA, international investment position.

Figure 2: Dollar Liabilities of Banks Headquartered outside the United States



Note: Data depict cross-border and local liabilities in all instruments vis à vis counterparty countries. Data exclude intragroup positions but include liabilities to other (unaffiliated) banks. From the end of 2015, data include positions reported by China and Russia (the latter up to the end of 2021). Data are in trillions of dollars.
Source: BIS consolidated banking statistics, BIS local banking statistics, and author's calculations

In this accounting example, changes in net foreign assets come from the current account and valuation. All amounts are USD billions.

$$\Delta NFA = \text{current account balance} + \text{valuation changes}.$$

Opening net foreign assets	−500
Current account deficit	−100
Net gain from asset prices and exchange rates	+150
Closing net foreign assets	−450

The external position improves by 50 despite a current account deficit of 100.

Source: BEA, international accounts methods.

Exorbitant Privilege and Exorbitant Duty

[Core Concept]

- **Funding advantage**: investors may accept a lower yield for the liquidity and collateral services of dollar assets.
- **Return differential**: the US may earn a different average return on external assets than it pays on liabilities. Differences in risk and valuation affect this comparison.
- **Exorbitant duty**: when risky foreign assets fall and safe dollar liabilities hold their value, the US external portfolio can lose wealth during global stress.
- Tabova and Warnock (2025) find that the positive differential in international *portfolio* returns disappeared in their data. Dollar assets can still provide liquidity and collateral services.

Source: Tabova and Warnock (2025), Exorbitant Changes in Three Parts.

Example: What Is a Funding Advantage Worth?

Suppose investors accept 4.0% on a liquid dollar bond, versus 4.5% on an otherwise comparable bond. Each issuer borrows USD 100B for one year.

$$\text{Annual interest saving} = 100\text{B}\,(4.5\% - 4.0\%) = \text{USD } 500\text{M}.$$

Foreign asset losses: The issuer invests the USD 100B in foreign equities. Their local price falls 20%, and their currency falls 10% against USD.

$$\text{Ending asset value} = 100(0.80)(0.90) = \text{USD } 72\text{B}.$$

Discuss: Compare the USD 500M funding saving with the USD 28B asset loss. What additional cash is needed for interest?

Global Dollar Liquidity and the Federal Reserve

- A non-US bank can own long-term dollar assets while funding them with short-term dollar borrowing or FX swaps. It must find dollars when that funding matures.
- If private lenders stop rolling over claims, selling assets can depress prices and weaken other banks holding the same assets.
- A Fed swap line supplies dollars to a partner central bank, which lends them to eligible banks against its own terms and collateral.
- The partner central bank bears the credit risk on its loans to banks. Dollar liquidity can bridge a funding shortage, while losses on bad assets still require capital or restructuring.

Source: BIS, foreign-currency funding measures (2008).

Safe-Asset Demand and the New Triffin Dilemma

[Core Concept]

New Triffin Dilemma: The US must supply safe dollar assets to meet growing world demand. Issuing more debt can strain its ability to service debt and undermine confidence in those assets; limiting issuance can leave the world short of safe assets.

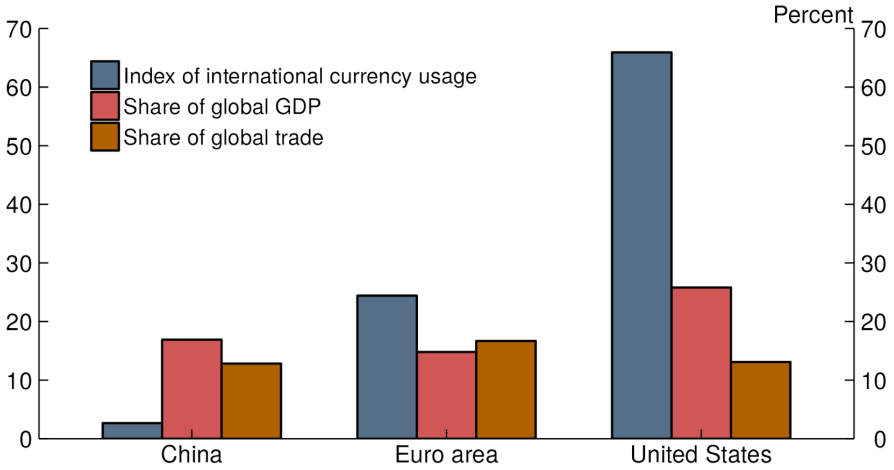
Illustration: Debt grows from USD 20T to 25T; annual revenue stays at USD 5T. Assume all debt is financed at the yield shown.

	Initial	More debt	More debt, higher yield
Debt (USD T)	20	25	25
Yield	3%	3%	4%
Annual interest (USD T)	0.60	0.75	1.00
Interest / revenue	12%	15%	20%

Discuss: How could doubts about repayment raise the cost of supplying safe assets?

Source: Emmanuel Farhi, Richmond Fed interview (2019). Numerical illustration is hypothetical.

Economic Size and International Currency Use



Source: Federal Reserve, CNY internationalization (2024).

What Would Support Greater EUR or CNY Use?

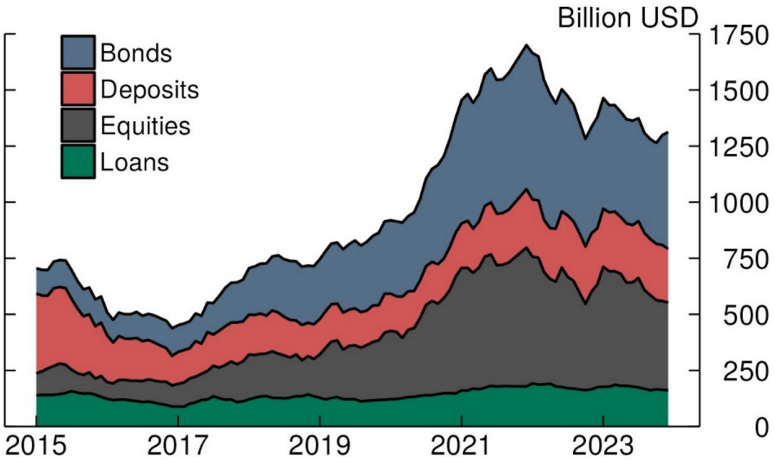
Consider the same four tasks for any proposed currency alternative.

1. Can firms invoice customers and reliably obtain the currency for debt service?
2. Can investors enter, hedge, and exit large positions at reasonable cost?
3. Is there a deep supply of liquid assets usable as collateral?
4. Will institutions support settlement and emergency liquidity during stress?

Euro markets face differences across sovereign issuers. CNY use depends partly on cross-border access and capital-account restrictions. Greater international use also requires deep asset markets and reliable access to funding.

Source: Federal Reserve, CNY internationalization (2024).

Foreign Holdings of Chinese Assets



Source: Federal Reserve, CNY internationalization (2024).

Exorbitant Privilege and Borrowing Costs

[Core Concept]

- Investors hold dollar assets to make payments, store cash, and provide collateral.
- Demand for safe, liquid dollar claims can raise their prices and lower their yields.
- This funding advantage is part of **exorbitant privilege**: the US can borrow in its own currency on favorable terms.
- Firms may benefit through lower benchmark yields. Their borrowing costs also depend on credit risk, maturity, and hedging.

If the advantage shrinks

Holding other drivers fixed, investors require higher yields. Existing bondholders lose value, and borrowers pay more on new or refinanced debt.

Source: Background: Federal Reserve, The International Role of the US Dollar (2025).

A Fall in Demand for Dollar Assets [Core Concept]

Scenario: investors reduce the share of their portfolios held in dollar assets.

- **Currency**: selling dollars for other currencies puts downward pressure on the dollar.
- **Bonds**: selling Treasuries puts downward pressure on their prices and upward pressure on their yields.
- **Funding**: higher market yields can raise the cost of issuing debt or refinancing a loan.
- **What investors buy matters**: a switch from long Treasuries to dollar bills mainly changes maturity exposure. A switch into euro assets also changes currency exposure.

A Weaker Dollar: Who Gains and Who Loses? [Core Concept]

Assume fixed dollar amounts and an appreciation of the firm's home currency.

Exposure	Immediate effect	Decision
Dollar sales receipts	Convert into less home currency	Hedge net receipts or revisit pricing
Dollar import payments	Cost less in home currency	Reassess purchasing costs and hedges
Dollar debt	Principal costs less in home currency	Check interest resets and refinancing
Dollar investments	Lose home-currency value through FX	Review currency hedge and asset risk

Net exposure matters: dollar receipts can offset dollar payments due at the same time. Existing hedges change the result.

Firm Example: Dollar Revenue and CNY Costs

Hypothetical exporter: in three months, it receives USD 10m from customers and pays USD 4m to suppliers. Its other costs are in CNY.

- Net dollar receipt: USD 10m − 4m = 6m.
- At 7.20 CNY per USD, the receipt is worth CNY 43.20m.
- At 6.60 CNY per USD, it is worth CNY 39.60m: a **CNY 3.60m loss** before hedging.
- Selling USD 6m forward locks the CNY value at the quoted forward rate. Hedging USD 10m would leave the supplier payment exposed.

Treasurer's decision

Match dollar inflows and outflows by date, then hedge the remaining exposure. If sales are uncertain, compare a partial hedge with an option and its premium.

Firm Example: A Smaller Funding Advantage

Hypothetical USD 20m refinancing, with annual interest and unchanged credit spread:

	Before	After
Dollar benchmark yield	4.0%	5.0%
Firm's credit spread	2.0%	2.0%
All-in borrowing rate	6.0%	7.0%
Annual dollar interest	USD 1.20m	USD 1.40m

- Annual interest rises by USD 200,000. A wider credit spread would add to the increase.
- Existing fixed-rate debt keeps its coupon until maturity. Floating-rate debt changes when its reference rate resets.

Decision: compare refinancing dates, fixed versus floating rates, and the return on projects that need new financing.

Choosing the Currency of a Loan

[Background Context]

- **Dollar revenue**: dollar debt can match cash inflows with debt service. Check that amounts and payment dates line up.
- **CNY revenue**: compare a CNY loan with a dollar loan hedged into CNY over the same maturity.
- Include the loan spread, FX hedge price, fees, and any collateral or rollover needs in the comparison.
- If dollar funding loses its cost advantage, local-currency borrowing may become more attractive. Use current all-in quotes to decide.

Connection to Weeks 4 and 8

The cheapest headline interest rate may come with expensive hedging. Leaving dollar debt unhedged creates a currency position that can gain or lose.

Hedge Fund: Matching a Trade to the Shock

Conditional trade ideas: each position needs a forecast relative to today's price.

View	Possible position	Main risk
Dollar weakens against EUR	Sell USD and buy EUR, or use an FX option	Dollar strengthens or option expires
Long Treasury yields rise	Reduce bond duration or short Treasury futures	Yields fall and bond prices rise
US assets remain attractive, but USD weakens	Keep assets and hedge dollar exposure	Asset prices fall despite the FX hedge

Financing matters: forward prices reflect interest-rate differences. Futures require margin. Options require a premium.

Fund Example: Bond Risk Plus Currency Risk

Hypothetical euro-based fund: USD 10m of Treasuries, modified duration 5, with no currency hedge. Ignore coupon income and financing over the shock window.

- Treasury yield rises by 1 percentage point. Approximate bond-price change: $-5 \times 0.01 = -5\%$.
- The euro value of one dollar falls by 8%.
- Euro portfolio return: $(1 - 0.05)(1 - 0.08) - 1 = -\mathbf{12.6}\%$.
- An FX hedge addresses the currency exposure. Shorter duration addresses sensitivity to yields. A hedge sized before the bond-price change may need adjustment.

Portfolio decision

Assess currency and duration separately. A dollar bond can lose through both channels at once.

A Dollar View Needs a Price and a Time Horizon

- A gradual decline in dollar demand may already be reflected in exchange rates and bond yields.
- A profitable position needs a further move relative to its entry price, after financing, hedging, and transaction costs.
- State the expected catalyst and holding period. Carry costs and margin calls can force an exit before a long-run view pays off.
- Use separate scenarios for lower dollar demand and a global scramble for dollar cash. The latter can strengthen the dollar and hurt a short-dollar position.

Fund manager's decision

Set the position size and cash buffer using the adverse scenario as well as the expected return.

April 2025: Investors Hedged Dollar Exposure

[Background Context]

- BIS analysis links part of the dollar's April 2025 decline to investors increasing hedges on their dollar assets.
- Investors can keep US bonds or equities while selling the associated dollar exposure forward.
- Dealers managing those forward positions can transmit the hedging pressure to the spot FX market.

Investment decision

The decision to own a US asset and the decision to hold its currency exposure can change separately.

Source: Shin, Wooldridge and Xia, US dollar's slide in April 2025: the role of FX hedging, BIS Bulletin 105 (20 June 2025).

Dollar Stablecoins and Currency Exposure

[Core Concept]

- A token designed to track USD 1 gives its holder dollar exposure. If the dollar falls against CNY, the token's CNY value falls even if its peg holds.
- An exporter receiving a dollar token still needs to manage the currency of its receipts and costs.
- If token issuers buy additional Treasury bills to meet new demand, this can support bill prices. Users switching from existing dollar assets can offset that demand.

Connection to the payment-routing game

Choose the payment route for its total cost and reliability. Choose the currency and hedge for the underlying cash flow.

Payment volume and token balances →

Cash Available When the Firm Needs It

[Core Concept]

- **Bank or currency change**: check whether customers can pay, suppliers accept the currency, and balances can be converted when bills are due.
- **Stablecoin route**: check who can redeem, how long redemption takes, and the fees for conversion into bank money.
- **Fund collateral**: allow for margin calls if bond prices fall or an FX position moves against the fund.
- **Contingency**: keep accessible cash and an alternative payment route for obligations that cannot wait.

Liquidity decision

A balance that holds its quoted value can still be unavailable when a payment is due.

Redemption example →

Hypothetical CNY-based firm, over the coming year:

- USD 10m customer receipts and USD 4m supplier payments arrive at year-end. Remaining operating costs are in CNY.
- USD 20m of debt is refinanced today for one year. Its annual rate rises from 6% to 7%.
- The year-end exchange rate is 6.60 CNY per USD, compared with a planning rate of 7.20. Assume the dollar cash flows stay fixed.

Discuss:

1. How much CNY value does the firm lose on its net operating receipt?
2. Does the weaker dollar offset the higher interest rate?
3. What should the treasurer compare before choosing the next loan and hedge?

Example: Cash Flows and Actions

[Background Context]

- **Operating receipt**: USD 6m falls from CNY 43.20m to CNY 39.60m. Loss: CNY 3.60m.
- **Annual interest**: the old plan was $20\text{m} \times 6\% \times 7.20 = \text{CNY } 8.64\text{m}$. The new outcome is $20\text{m} \times 7\% \times 6.60 = \text{CNY } 9.24\text{m}$.
- Interest costs rise by CNY 0.60m despite the weaker dollar. Cash after operating flows and interest falls by **CNY 4.20m**, before other costs and principal repayment.
- At year-end the USD 20m principal also falls in CNY value, from 144m to 132m. Its repayment or refinancing needs a separate funding plan.

Action: compare a dollar loan matched to receipts with a hedged dollar loan and a CNY loan. Include interest and principal in the dated exposure schedule.

A Dollar Shock and Asset Prices: The Joint Distribution [Core Concept]

A euro investor holds a USD asset. Let g be the change in EUR per USD and r the asset's USD return over one shock window. Ignore income, costs, and financing.

$$R_{\mathrm{EUR}} = (1 + r)(1 + g) - 1 = r + g + rg.$$

State	Probability	USD FX g	Treasury r	US equity r
Calm	60%	0%	1%	8%
Global flight to safety	25%	10%	5%	−20%
Loss of confidence in USD assets	15%	−10%	−8%	−15%

Use these hypothetical probabilities for the following exercises.

Exercise: Returns in the Investor's Currency

[Background Context]

Use the preceding three-state distribution. Initial wealth is EUR 1M, invested entirely in either the Treasury portfolio or the US equity portfolio.

1. Calculate the EUR return on each portfolio in every state.
2. Calculate each portfolio's expected EUR return and probability of a loss.
3. For a portfolio with initial weights of 60% Treasuries and 40% equities, calculate the return in each state and its expected return. Assume no interim rebalancing.
4. Which shock makes both the dollar and Treasury prices fall? Why would a portfolio built only around flight to safety miss this risk?

Solution: Dollar and Asset Returns Compound

EUR return	Treasuries	Equities	60/40 portfolio
Calm	1.0%	8.0%	3.8%
Flight to safety	15.5%	−12.0%	4.5%
USD confidence loss	−17.2%	−23.5%	−19.72%
Expected return	1.895%	−1.725%	0.447%
Probability of a loss	15%	40%	15%

$$E[R_{T,\mathrm{EUR}}] = 0.60(1\%) + 0.25(15.5\%) + 0.15(-17.2\%) = 1.895\%.$$

The 60/40 portfolio has expected ending wealth EUR 1,004,470. Its confidence-loss outcome is EUR 802,800.

Why Averaging the Two Returns is Incorrect

The exact expected home-currency return includes the interaction of asset and FX returns.

$$E[R_{\mathrm{EUR}}] = E[r] + E[g] + E[rg].$$

For the Treasury portfolio:

$$E[r] = 0.65\%, \qquad E[g] = 1.00\%, \qquad E[rg] = 0.245\%.$$

$$E[R_{\mathrm{EUR}}] = 0.65\% + 1.00\% + 0.245\% = 1.895\%.$$

- Multiplying mean gross returns gives $1.0065(1.01) - 1 = 1.6565\%$, which misses their covariance.
- The difference is $\mathrm{Cov}(r, g) = 0.2385$ percentage points in expected return. Treasuries and USD rise together in the flight-to-safety state.

Exercise: A Hedge on the Initial USD Principal

Keep the Treasury portfolio and the same distribution. Sell the initial USD principal forward. Assume the forward rate equals the initial spot rate and the contract is cash-settled in EUR at the end. Ignore fees, collateral, and default.

$$R_{\text{hedged}} = r + rg, \qquad \text{forward payoff / initial EUR wealth} = -g.$$

1. Calculate the hedged return in each state and its expected return.
2. In which state must the investor pay cash on the forward? How much for an initial EUR 1M investment?
3. Compare the hedged and unhedged portfolios' worst state. Why does the hedge leave some FX exposure?

Solution: The Hedge Changes Both Wealth and Cash Needs

State	Unhedged	Forward contribution	Hedged
Calm	1.0%	0%	1.0%
Flight to safety	15.5%	−10%	5.5%
USD confidence loss	−17.2%	10%	−7.2%
Expected return	1.895%	−1.0%	0.895%

- In flight to safety, the forward requires EUR 100,000. The Treasury holding has gained value, but the cash must be available on the settlement date.
- The confidence-loss outcome improves from −17.2% to −7.2%. The initial-principal hedge leaves the term rg exposed.

Exercise: Exorbitant Duty in a Two-State Model [Core Concept]

A simplified US external portfolio holds foreign equities worth USD 100B and owes fixed USD liabilities worth 80B. Liabilities keep their USD value. There are no new financial flows, income payments, or other positions.

State	Probability	Foreign equity local return	USD per foreign currency
Expansion	80%	10%	Rises 5%
Global crisis	20%	−30%	Falls 10%

1. Initial net foreign assets are USD 20B. Find ending net foreign assets in both states and their expected value.
2. Explain how a positive average valuation change can coexist with a large loss in the crisis state.

Solution: Average Gains and Crisis Losses [Core Concept]

USD billions	Expansion	Global crisis	Expected value
External asset value	$100(1.10)(1.05) = 115.5$	$100(0.70)(0.90) = 63$	105.0
External liabilities	80	80	80
Net foreign assets	35.5	-17.0	25.0
Change from initial 20	15.5	-37.0	5.0

- Expected valuation gain is USD 5B, but the global crisis produces a USD 37B loss. The foreign holders of the assumed fixed dollar claims retain their dollar value.
- US equity and direct-investment liabilities also change value, so a full calculation includes valuation changes on both sides.

A Treasury Decision under Correlated Dollar Shocks [Core Concept]

A firm with CNY costs has USD 6M net export receipts after dollar supplier payments and refinances USD 20M debt. The table gives three possible outcomes for annual cash after interest. The debt principal is refinanced.

State	Probability	CNY per USD	New USD rate	CNY cash after interest
Calm	60%	7.2	6%	$(6-1.2)7.2=34.56$M
Dollar demand loss	25%	6.6	7%	$(6-1.4)6.6=30.36$M
Global funding squeeze	15%	7.8	9%	$(6-1.8)7.8=32.76$M

1. Calculate expected cash and the probability that cash falls below CNY 32M.
2. Which exposure would a forward hedge cover? Which exposure requires a rate or refinancing decision?

Treasury Distribution: Expected Cash and the Shortfall

$$E[\text{CNY cash}] = 0.60(34.56) + 0.25(30.36) + 0.15(32.76) = 33.24\text{M}.$$

State	Cash relative to CNY 32M requirement	Probability
Calm	Surplus 2.56M	60%
Dollar demand loss	Shortfall 1.64M	25%
Global funding squeeze	Surplus 0.76M	15%

- The expected cash exceeds 32M, but the firm still has a 25% shortfall probability.
- An FX hedge fixes the conversion of a specified dollar amount. A fixed-rate loan or interest-rate hedge addresses interest changes. Treasury must also arrange the refinancing.

Week 10 Summary: Decisions for Firms and Funds

- **Payments**: compare fees, conversion costs, prefunding, and the risk of delay or loss.
- **Firms**: map net cash flows by currency and date. Reassess margins, refinancing, and hedges after a dollar shock.
- **Exorbitant privilege**: a smaller funding advantage can mean higher required yields. Existing bondholders and new borrowers face different consequences.
- **Funds**: separate currency, interest-rate, and funding risk. Judge a trade against today's price and the cash needed to survive adverse moves.
- **Digital money**: changing the payment technology can leave the currency exposure in place.

Next week: emerging-market crises, sudden stops, and sovereign debt.

Appendix: The Ledger Economics Trilemma

Abadi and Brunnermeier (2022) show that, in their model, no ledger has all three properties below.

1. **Self-sufficient (trustless)**: needs no outside power to punish record-keepers.
2. **Rent-free**: pays record-keepers no economic rents (returns above the cost of providing the service).
3. **Resource-efficient**: wastes no resources such as computing power.

Examples:

- Bitcoin's proof of work pays validators and uses large computing resources. Ethereum now uses proof of stake, which uses less computing power but has other incentive and governance costs.
- A commercial-bank ledger uses few resources. The bank charges fees, and users must trust the bank and its supervisor.
- A central bank ledger uses few resources and charges low fees. Users must trust the central bank.

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Appendix: Domestic Fast Payment Systems in Detail

- **Brazil Pix** (2020): the central bank operates the scheme. Large institutions must participate, which gave Pix wide reach. Fees depend on the user and the provider.
- **India UPI**: NPCI operates one instant payment scheme for all regulated banks and payment apps. Common standards and wide smartphone use support high usage.
- **US ACH and FedNow**: ACH settles batches, with same-day and next-day options. FedNow (2023) offers 24x7 instant settlement among participating institutions.
- **Hong Kong FPS** (2018): supports aliases and QR payments, with selected cross-border links.

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Appendix: Nostro and Vostro Accounts

A Hong Kong bank pays a beneficiary bank abroad through a US correspondent. The same account has two names depending on whose books you read.

HK Bank's books

- **Nostro asset**: USD balance held at the US correspondent, +$1,000,000.
- Here the balance is prefunded. A bank can also use netting or a credit line.

US Correspondent's books

- **Vostro liability**: the HK bank's account on its books, +$1,000,000.
- On instruction, it debits the vostro account and credits the beneficiary bank.

The economic cost is the balance times the spread between the bank's funding rate and the interest the nostro earns. One balance supports many payments.

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Appendix: Separate Three-Route Cost Illustration

[Background Context]

Principal: USD 1M. Carry: \$100 per day per \$1M committed. One bp is 0.01%. All charges are paid separately by the sender.

Cost (USD)	A	B	C
Fixed fee	150	400	50
FX spread cost	$1\text{M}(0.0008)$	$1\text{M}(0.0004)$	$1\text{M}(0.0003 + 0.0003)$
Committed cash (USD M)	1	0.2	1
Carry cost	$1(2)(100) = 200$	$0.2(1)(100) = 20$	$1(0.05)(100) = 5$
Risk charge	$1\text{M}(0.00025)(2) = 500$	0 (PvP)	$1\text{M}(0.0001) = 100$
Total	1,650	820	755

FX cost = principal $\times$ spread. Carry = committed USD M $\times$ days $\times$\$100. C uses two 3 bp legs on the same principal (a small-spread approximation), a 1 bp redemption charge and about one hour of committed cash.

The risk charges are assumed prices. Prefunding time and the settlement gap are separate inputs.

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Bargaining in a dealer network (Duffie and Wang, 2026)

- Dealers in a chain of trades (a network $G = (N, E)$, with dealers N and trading links E) bargain in pairs over asset amounts and cash payments.
- Without atomic settlement, a dealer that completes leg 1 first has a weaker position on leg 2. The next dealer gains extra profit from this bargaining advantage: a holdup rent.

The greenlight protocol

- Each participant sends a conditional approval (a "greenlight") for the proposed terms of each bilateral contract.
- A smart contract makes all approved contracts binding **only if** every participant approves at the same time (all or none).
- **Model result**: under the paper's assumptions, the protocol removes the holdup and gives the efficient allocation.

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Appendix: Stablecoin Redemption and Cash Needs

Hypothetical firm: USD 1m of dollar tokens and a USD 1m supplier payment due today.

- The firm cannot redeem directly with the issuer. It can sell tokens on an exchange at USD 0.99 each.
- Selling today raises USD 990,000 before fees, leaving a USD 10,000 shortfall.
- An eligible intermediary can redeem at par. Its fees, capacity, and settlement time set the price it offers the firm.
- Waiting for the peg to recover leaves the supplier payment unfunded today. The firm needs another cash balance or a credit line.

Treasurer's decision

Check which redemption rights your firm has, and how long it takes to get bank money.

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US Payment Rails: Separate Operator, Rules, and Settlement

- **ACH:** credit and debit transfers in batches, with same-day options. Two operators run ACH: FedACH (Federal Reserve) and EPN (The Clearing House). Nacha sets the network rules.
- **Fedwire:** the Federal Reserve's real-time gross settlement system for eligible institutions. Banks use it for large, urgent transfers.
- **RTP and FedNow:** The Clearing House launched RTP in 2017. The Federal Reserve launched FedNow in July 2023. Both settle instant payments between banks, 24 hours a day.
- **Interface and rail:** the banking app is the interface. The rail sets the message format, the operating hours, the settlement method, and the rules for participants.

The systems settle in different ways. Fedwire and FedNow settle each payment gross. ACH settles net amounts in batches. RTP settles from a prefunded joint account at the Federal Reserve.

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Brazil Pix: Design and Merchant Costs

[Background Context]

- Pix launched in November 2020 under Banco Central do Brasil. Participation was mandatory for institutions with more than 500,000 active customer transaction accounts.
- Users start transfers with aliases or QR codes. One common standard lets users of different providers pay each other.
- A 2022 BIS review reported a 2021 merchant cost of about 0.22% for Pix, compared with 2.2% for credit cards.
- Example: on a BRL 1,000 sale, the merchant pays BRL 2.20 with Pix and BRL 22.00 with a credit card. A credit card also gives the buyer credit and chargeback rights.

Source: BIS Bulletin 52, 2022.

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Hong Kong FPS and India UPI

Hong Kong FPS

FPS launched in 2018. It settles HKD and CNH payments between banks and stored-value facilities. Users can pay to a phone number, an email address, an FPS ID, or a QR code. The FPS–PromptPay QR link (4 December 2023) lets participating users pay merchants in the other economy.

India UPI

NPCI's UPI connects banks and payment apps in India. Apps such as PhonePe, Google Pay, and Paytm are interfaces. A payment between two apps settles between the users' banks. Common addresses and QR standards help network effects.

Adoption depends on access, price, merchant acceptance, trust, and policy.

Source: HKMA, November 2023 briefing.

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Smart-Contract Escrow: A Shipment Example

Example: an importer agrees to pay 100 token units when the exporter supplies an accepted shipping document.

1. The buyer places 100 units in an escrow contract. The contract locks the funds until it releases or refunds them under agreed rules.
2. An approved data provider sends a verified shipping record. The contract checks the agreed conditions.
3. If the conditions hold, the contract pays 100 to the exporter. If not, the funds stay locked until the dispute or time-limit rules apply.
4. The code cannot inspect the goods. False documents, a hacked data provider, legal disputes, or code errors can cause losses.

Compare with a letter of credit (LC). Both make payment depend on documents. In an LC, a bank is legally bound to pay. In the escrow, the buyer prefunds and the code pays. The ledger design sets who can read and write each transaction. A permissioned ledger can keep transactions private.

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Cross-Border CBDC Experiments: The Original Cases

Project	Participants	What was tested
Dunbar (2022)	Australia, Malaysia, Singapore, South Africa	Prototypes for shared settlement using multiple wholesale CBDCs.
Icebreaker (2023)	Israel, Norway, Sweden	A hub connecting domestic retail-CBDC systems, with FX providers.
mBridge (2024)	China, Hong Kong, Thailand, UAE. Saudi Arabia joined in 2024	A shared DLT platform reached minimum viable product stage. The BIS transferred the project to the members in October 2024.

Source: BIS, mBridge.

Source: BIS, Dunbar.

Source: BIS, Icebreaker.

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Wallets and Stablecoins: Trace the User's Claim

[Background Context]

- **Platform wallets:** Alibaba built Alipay for its online shops. Tencent built WeChat Pay into its messaging app. A platform that combines payments, shopping, and messaging gets new customers at low cost.
- A wallet can hold stored value, show a linked bank deposit, or start a card payment. Each gives the user a different legal claim. Example: Apple Pay starts a card payment, and Octopus holds stored value.
- When a user buys a stablecoin from the issuer, bank money becomes a token. Redemption reverses this step. On an exchange, holders trade tokens with each other.
- Issuers and wallets earn income from reserve assets, fees, and other financial services. Rules on reserve segregation decide who bears losses if the issuer fails.

Always ask: who owes the user money, what backs that obligation, and where does final settlement occur?

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Separate Illustration: Increasing the Payment Time to Five Days

Baseline costs are Route A \$1,650 and Route B \$820. For Route A, the scenario increases prefunding from 2 to 5 days and the settlement gap from 2 to 5 days. Fees and FX spread are unchanged.

1. Extra prefunding cost: $(5-2)$ days $\times$ \$100/day $=$ \$300.
2. Extra settlement-risk charge: \$1M $\times$ 0.00025 $\times$ $(5-2) =$ \$750.
3. New Route A total: $1{,}650 + 300 + 750 =$ \$2,700.
4. If Route B stays at \$820, its advantage increases from \$830 to \$1,880. Check if the same shock also changes Route B.

Decision rule: for each feasible route, compare the full expected cost, the time, the finality, and the risk. A route with a low fee can have a high total cost.

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Dollar Invoicing: The Buyer's Currency Bill Changes

[Background Context]

A Vietnamese exporter invoices a Korean customer USD 100. The dollar invoice price is fixed for this order. The buyer has no hedge.

1. At 1,300 KRW per USD, the buyer's cost is $100 \times 1{,}300 =$ KRW 130,000.
2. At 1,430 KRW per USD, the same invoice costs KRW 143,000, an increase of KRW 13,000 or 10%.
3. The buyer can buy less in future or accept a lower margin. Contracts, inventories, and local prices set how fast the buyer passes on the cost.
4. The exporter still receives USD 100. Its VND receipt depends on the USD/VND rate and its hedges.

Dollar invoicing moves changes in the dollar exchange rate into trade between two non-US countries.

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Stablecoin Payment Flows and Reserve Stocks

[Background Context]

Hypothetical annual trade flow: CNY 2 trillion. Suppose 20% is paid through a stablecoin whose average balance turns over 20 times per year.

1. Annual stablecoin payment flow is $2\text{T} \times 20\% = \text{CNY } 400\text{B}$.
2. From $V = vM$, average token stock is $M = 400/20 = \text{CNY } 20\text{B}$ equivalent.
3. At 100% reserve backing, required reserves for that stock are 20B equivalent, not the full 400B payment flow.
4. At turnover 10, the same flow requires 40B. Extra balances that users keep for safety increase the stock further.

Net new demand for US assets is smaller when users move out of existing dollar deposits or other dollar instruments into tokens.

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Appendix: Why Exporters Keep Prices in Dollars

[Background Context]

Gopinath (August 2026) gives these reasons why firms invoice in dollars.

- **Dollar-priced inputs:** exporters are also importers. When their inputs are priced in dollars, part of their cost is in dollars, so a depreciation of their own currency lowers their dollar costs only a little.
- **Global value chains:** the domestic value-added share of exports has declined. Imported inputs are a larger share of each export's cost, and suppliers and competitors also price in dollars.
- **Evidence across goods:** firms that rely more on imported inputs price more of their exports in dollars.
- **Contracts:** contracts typically fix a price and allow a range of quantities. The contracted price therefore affects how much is bought until the contract is repriced.

With fully flexible prices the invoicing currency would not matter. It matters because prices stay fixed for the length of a contract.

Source: Gita Gopinath, Conversations with Tyler, Ep. 291, recorded 27 August 2026.

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Appendix: A Depreciation versus an Import Tariff under Dollar Pricing [Background Context]

Illustration: a Thai exporter sells to a Japanese importer at USD 100, fixed in dollars for the contract. Unit cost is USD 50 of dollar-priced inputs plus THB 1,400 of local costs, at 35.0 THB per USD.

1. Initial unit cost: $50 + 1{,}400/35.0 =$ USD 90. The margin is USD 10.
2. The price of a dollar rises 10%, to 38.5 THB. Unit cost falls to $50 + 1{,}400/38.5 =$ USD 86.36, only 4% lower. The importer still pays USD 100: no pass-through while the price is fixed.
3. Instead, a hypothetical 25% import tariff with the price kept at USD 100: the importer pays $100 \times 1.25 =$ USD 125, full pass-through at the border.
4. Absorbing the tariff would require a price of $100/1.25 =$ USD 80, below unit cost.

Evidence on pass-through

Cavallo, Gopinath, Neiman, and Tang (2021) find near-complete pass-through of import tariffs to border prices in their sample, while exporters barely cut dollar prices when the dollar appreciated. Retail prices rose less because retailers absorbed part of the tariff.

Source: Cavallo, Gopinath, Neiman, and Tang, AER: Insights, 2021.

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